IOWA STATE UNIVERSITY
Department of Chemical and Biological Engineering

Undergraduate Research

Colocalization-Confirmed Transcriptome-Wide Association Study of Ankylosing Spondylitis to Prioritize Tissue-Specific Drug Targets

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Introduction

Ankylosing spondylitis (AS) is a chronic inflammatory arthritis of the spine and sacroiliac joints affecting 0.1–0.5% of the population, with onset typically before age 30. GWAS have identified over 100 risk loci, but most fall in non-coding regions.

Because many AS-associated variants do not directly implicate a protein-coding gene, linking disease risk to *tissue-specific gene regulation* is an important step toward identifying biologically meaningful mechanisms and therapeutic targets.

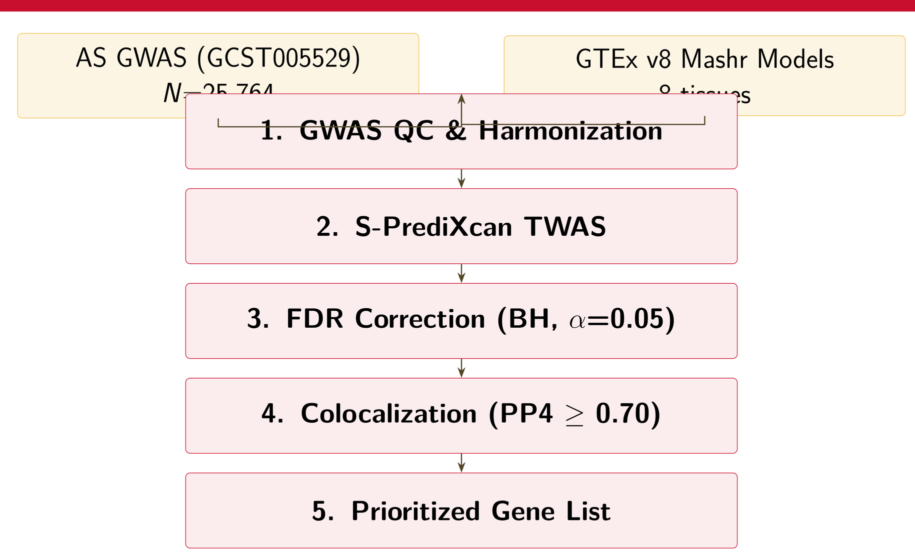
Transcriptome-wide association studies (TWAS) integrate GWAS with gene expression models to identify genes whose genetically regulated expression is associated with disease. However, TWAS hits can arise from *linkage disequilibrium (LD) contamination* rather than true shared causal variants.

Bayesian colocalization tests whether a GWAS signal and an eQTL signal share the same causal variant, helping distinguish high-confidence regulatory mechanisms from LD-driven associations.

Key Question

Which genes have genetically regulated expression that **causally colocalizes** with AS risk variants, and in which tissues?

Methods



GWAS QC: 123,941 SNPs → liftOver hg38 → 14,032 palindromic removed → 109,893 harmonized.

Tissues: Whole Blood, Spleen, Lung, Colon Sigmoid, Colon Transverse, Small Intestine, Muscle Skeletal, Adipose Subcutaneous.

Discussion

- *ERAP1* (MHC-I antigen processing) is a known AS gene, validating our pipeline. Colocalizes in Whole Blood and Small Intestine.
- Novel candidates *GPR25*, *CCDC116*, *NPIPB7* warrant experimental follow-up as tissue-specific drug targets.
- *CARD9* and *FAS* implicate innate immune and apoptotic pathways beyond HLA-B27.
- Colocalization reduced 248 TWAS hits to 35 confirmed pairs, filtering LD-driven false positives.

TWAS Results

S-PrediXcan tested **2,316 gene-tissue pairs** across 8 GTEx v8 tissues. After Benjamini–Hochberg FDR correction ($\alpha = 0.05$), **248 significant associations** were identified, spanning genes on multiple chromosomes.

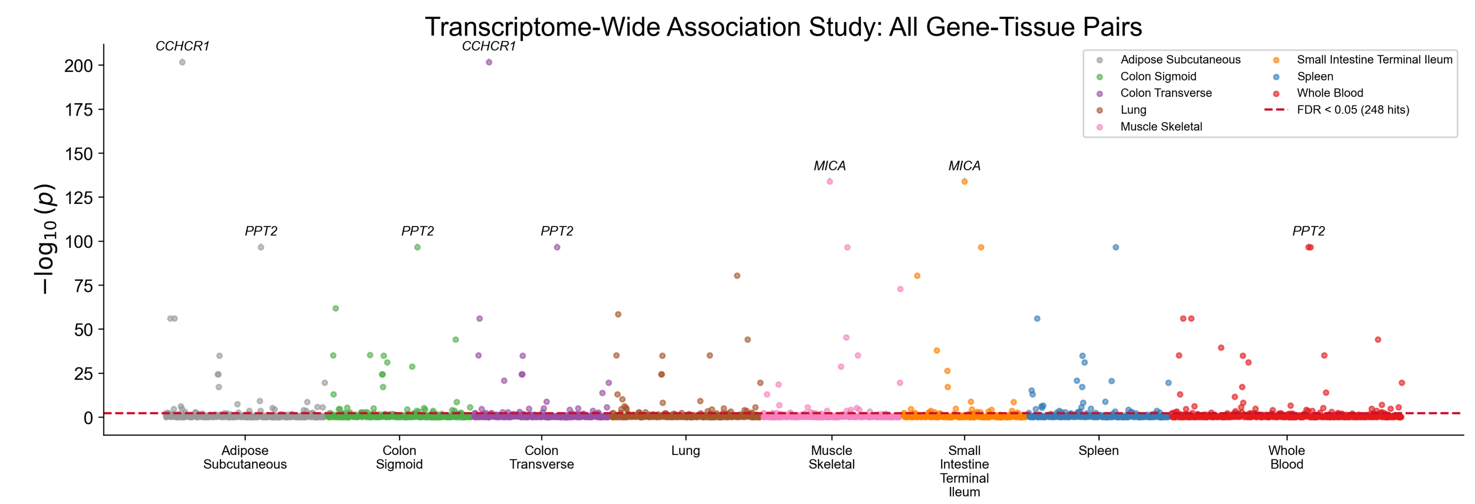


Figure. Manhattan-style plot of TWAS $-\log_{10}(p)$ across 8 tissues. Top gene symbols annotated. Dashed line: FDR < 0.05.

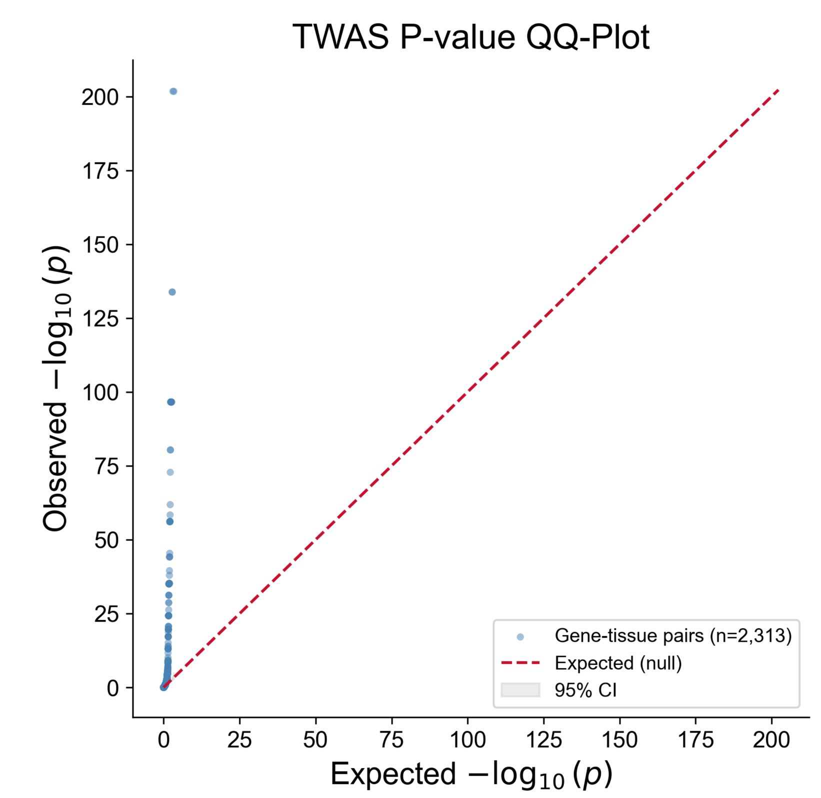


Figure. QQ-plot: departure from null consistent with polygenic architecture. Grey band: 95% CI.

Colocalization Results

Of the 248 TWAS hits, Bayesian colocalization (coloc.abf) identified **35 gene-tissue pairs (18 unique genes)** with PP4 $\geq$ 0.70, confirming a shared causal variant between GWAS and eQTL signals.

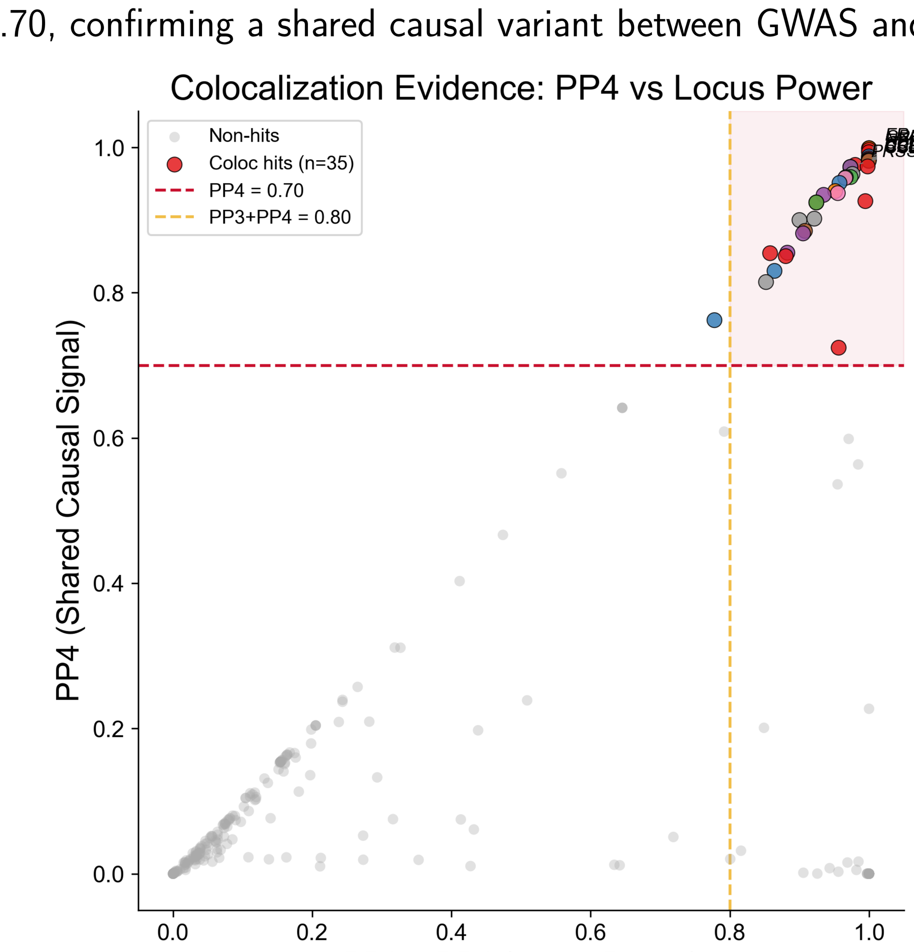


Figure. PP4 vs. locus power (PP3+PP4). Upper-right: colocalization-confirmed hits.

Conclusion

Integrating S-PrediXcan TWAS with Bayesian colocalization identified **18 unique genes** with shared causal variants at AS risk loci, prioritizing tissue-specific targets for therapeutic development.

This framework improves interpretability by narrowing a broad set of TWAS signals to a smaller group of higher-confidence gene-tissue pairs, providing a focused set of candidates for downstream fine-mapping and functional validation in AS.

Future Directions

- Weighted protein interaction network of prioritized genes via Seq2Bind [1,2]
- Fine-mapping (SuSiE) and validation in independent AS cohorts
- Drug repurposing analysis for top targets

Top Colocalized Genes

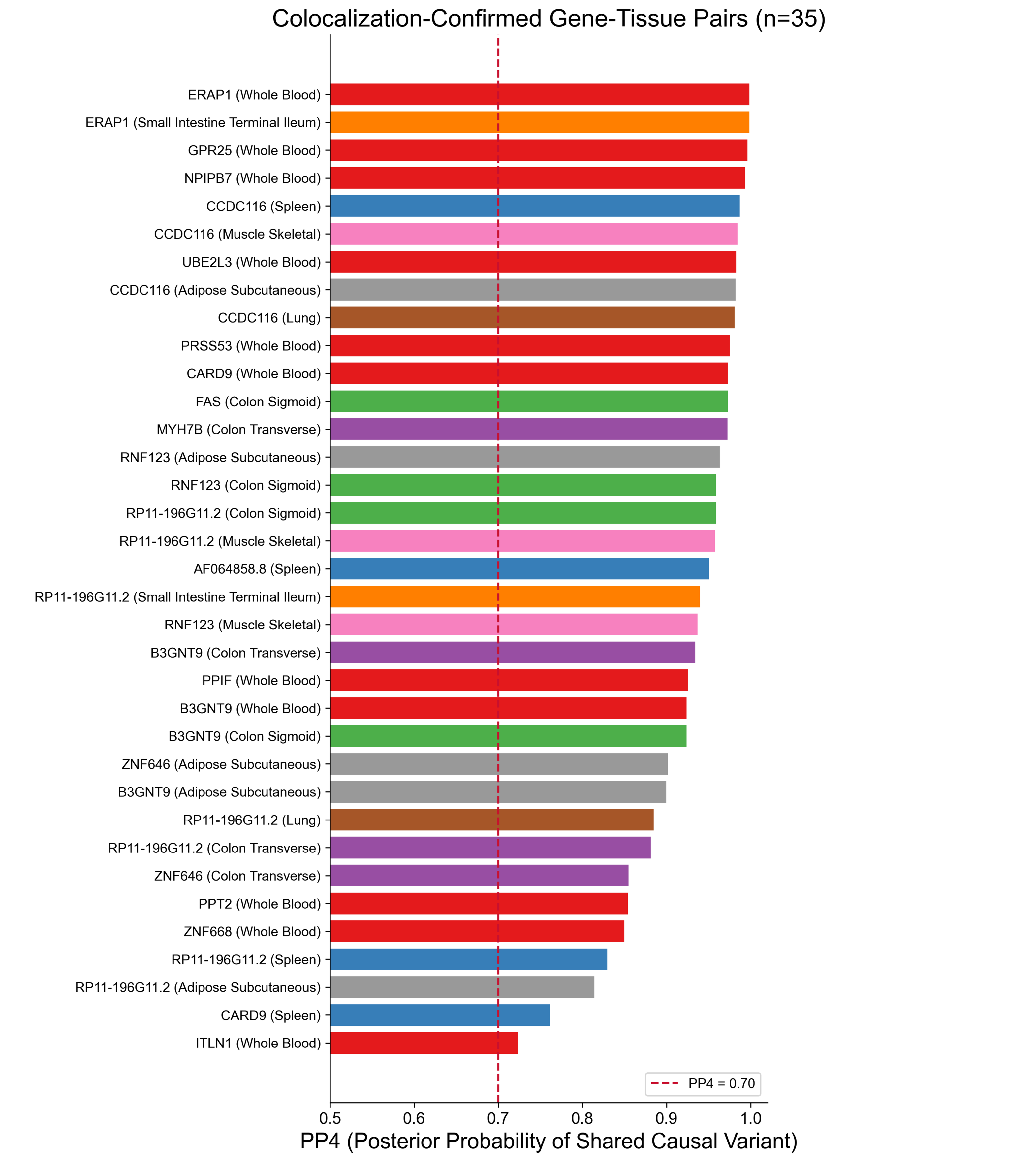


Figure. All 35 coloc-confirmed gene-tissue pairs ranked by PP4. Color indicates tissue. Dashed line: PP4 = 0.70.

Gene	Tissue	PP4	z	FDR p
ERAP1	Whole Blood	0.999	13.28	2.5e-38
GPR25	Whole Blood	0.997	−7.03	6.0e-11
NPIPB7	Whole Blood	0.993	−5.16	5.9e-06
CCDC116	Spleen	0.988	−5.16	5.9e-06
UBE2L3	Whole Blood	0.983	5.32	2.5e-06
CARD9	Whole Blood	0.974	4.94	1.7e-05
FAS	Colon Sigmoid	0.973	4.42	1.9e-04

Selected References

1. Ma X, Dey S, et al. Seq2Bind webserver. *NAR Genom Bioinform.* 2025;7:lqaf154.
2. Dey S, Chowdhury R. Twin Peaks. *arXiv*:2509.22950. 2025.
3. Barbeira AN, et al. S-PrediXcan. *Nat Commun.* 2018;9:1825.
4. Giambartolomei C, et al. coloc.abf. *PLoS Genet.* 2014;10:e1004383.

Acknowledgements & Code

Thanks to **Dr. Ratul Chowdhury** for mentorship. GWAS data from NHGRI-EBI GWAS Catalog (GCST005529). Expression models from GTEx Consortium (v8). S-PrediXcan and coloc.abf software authors.

github.com/yellepeddibk/as-twas-coloc